NovaNest Webhook Call Log Pipeline

Retell AI Lesson 5: Webhooks, Post-Call Events & Automation Pipelines
Project Summary

GitHub Repository	https://github.com/shaikshahid777/novanest-webhook-pipeline
Demo Video (Loom)	https://www.loom.com/share/75985b0632364804840d82b19f987200
Stack	Retell AI, n8n Cloud, Google Sheets
Agent	NovaNest Consultation Booking Agent

1. Project Overview

This project is an event-driven automation that receives Retell AI webhook events from the NovaNest Consultation Booking Agent and writes one structured row per completed call into a Google Sheet call log. The webhook is registered at agent level, and the **call_analyzed** event is the source of record, because it carries the call summary and the Post-Call Analysis fields.

Flow: **Retell call ends** → **webhook to n8n** → **event filter** → **duplicate check** → **field mapping** → **Google Sheet row.**

2. Workflow Steps (n8n)

#	Node	Behavior
1	Webhook	Receives POST from Retell and returns 200 OK immediately (response-first pattern).
2	Is call_analyzed?	True continues. call_started, call_ended and other events go to "Discarded (other event)" (acknowledged, no write).
3	Has call_id?	Missing call.call_id goes to a Stop and Error node, visible as a failed execution.
4	Lookup call_id in Sheet	Idempotency check against the call_id column.
5	Duplicate?	If found, path "duplicate skipped" and nothing is written.
6	Map & Transform	Code node extracts and transforms fields (see section 3).
7	Append Row	Writes one row. Retry On Fail enabled, 3 tries (2 retries).

3. Field Mapping

Column	JSON path (under call)	Transformation
call_id	call_id	Idempotency key
call_date	start_timestamp	YYYY-MM-DD, America/Chicago
call_time	start_timestamp	HH:MM CT
duration_seconds	duration_ms	Divide by 1000, round
call_status	call_status	As received
caller_name	call_analysis.caller_first_name + caller_last_name	Joined with a space
caller_phone / caller_email	call_analysis.*	N/A if null
qualification_status	call_analysis.qualification_status	Hot / Warm / Cold / Disqualified, N/A if null
call_summary	call_analysis.call_summary	N/A if null

Column	JSON path (under call)	Transformation
recording_url	recording_url	Valid https URL kept; otherwise "URL expired"
follow_up_status	(default)	Always "Pending"

Custom Post-Call Analysis fields are read from call.call_analysis merged with call.call_analysis.custom_analysis_data.

4. Test Results

ID	Test	Result	Evidence
TC-001	call_analyzed creates a row	Pass	Real web call (Mia Torres, Hot) produced the row
TC-002	call_started / call_ended discarded	Pass	Real call: both events took the Discarded path; also tested by payload
TC-003	Duplicate prevention	Pass	Same payload sent twice; second run took "duplicate skipped", no new row
TC-004	Field mapping accuracy	Pass	Real call output and sheet row checked column by column
TC-005	Recording URL expiry	Pass	Empty recording_url gave "URL expired"
TC-006	Missing call_analysis gives N/A	Pass	Payload without call_analysis gave N/A in affected columns
TC-007	200 OK response speed	Pass	All deliveries acknowledged in about 1-4 seconds, well under 10 s
TC-008	Distinct rows per call	Pass	Rows for Mia (real call), Ravi Kumar (Warm) and Sara Lee (Disqualified, via payload)
TC-009	follow_up_status = Pending	Pass	Pending on every row
TC-010	Failed execution visible	Pass	Missing call_id produced a red failed execution with a clear error message

One full end-to-end real Retell call was run; the remaining scenarios were verified with payload-injection tests (curl.exe). Additional real calls were skipped to save call credits.

5. Error Handling

- Missing call_id: no row written; failed execution with message "Payload missing call.call_id - row not written".
- Missing or null call_analysis fields: row written with N/A.
- Expired or invalid recording URL: "URL expired" placeholder.
- Sheet write failure: automatic retry (3 tries), then failed execution in n8n Executions.

6. Assumptions

- Post-Call Analysis from earlier topics is configured on the agent.
- Times are shown in Central Time (America/Chicago).
- n8n Cloud provides the public HTTPS webhook URL.

7. Limitations

- recording_url is validated but not re-uploaded, so the link expires after about 10 minutes.

- Very rapid concurrent deliveries within a few seconds can overwrite rows in Google Sheets (seen in a curl burst test); real calls arrive minutes apart. Lookup then append is not atomic.
- In the real call, the Topic 7 create_booking_request function returned an error. This is outside this topic and did not affect the pipeline, which still logged the call correctly.

8. Enhancements

- Download and store the recording permanently (Google Drive or S3) immediately on receipt.
- Verify the Retell webhook signature.
- Slack or email alert for Hot leads.
- Dead-letter tab and a global n8n error workflow.
- Use a CRM or Airtable with call_id upsert for atomic idempotency.